

Nikunj Pradhan

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EDUCATION

Missouri University of Science and Technology

Bachelor of Science, Computer Science

GPA: 4.0/4.0

Aug 2023 - Dec 2026

EXPERIENCE

Data Engineer Co-Op

Hunter Engineering Company

Jan 2025 – Aug 2025

Bridgeton, MO

- Engineered and scaled full-stack applications (FastAPI, React, SQL Server) providing real-time dashboards, data pipelines, and annotation workflows for engineering teams.
- Automated benchmarking workflows by engineering Python CLI tools across AWS and Azure ML pipelines, enabling comparison with vendor models and streamlining team workflow.
- Accelerated distributed multi-GPU training workflows by 66% and development-to-production cycles through close collaboration with R&D teams.
- Created and maintained custom ETL pipelines using FastAPI and Python to clean, validate, and process metadata and image data from 600+ worksites, ingesting over 2M images into Microsoft SQL Server and AWS S3.
- Built a CNN-based regression model predicting object coordinates, improving geometric correction accuracy by 23%.

Backend Engineer

Tally Receipts

June 2025 – July 2025

St. Louis, MO

- Redesigned and migrated backend from MongoDB to AWS DynamoDB, redesigning schemas for efficient querying, scalability, and data lifecycle management.
- Built LLM-powered receipt categorization service, reducing manual entry effort for users and integrating seamlessly with calendar workflows (Google, Microsoft).
- Integrated FidelSDK for secure card linking in the frontend, designed business logic for safe card tracking in the database, and implemented webhooks to collect user purchase receipts.
- Refactored and modernized Node.js + Express.js services and integrated AWS Lambda, Amplify, and IAM, streamlining backend logic and infrastructure.

Computer Vision Researcher

Missouri University of Science and Technology

Aug 2023 – Dec 2024

Rolla, MO

- Improved CV model accuracy by 62% through optimization, hyperparameter tuning, and algorithmic refinement using C++ OpenCV and Point Cloud.
- Developed a MATLAB/C++ application simulating 3D triaxial shear tests with deep learning and parallel computing for soil compression modeling, incorporating advanced CV techniques like watershed segmentation and 3D model triangulation via ray tracing.

PROJECTS

Devanagari OCR Pipeline Application | FastAPI, Python, Pytorch, NLTK, MLOps

- Designed a full Nepali OCR pipeline +130 characters CRNN text recognizer with integrated real-time line-detection trained on 500k+ samples and delivered as a FastAPI inference workflow.

Movie Recommendation & Watchlist Platform | FastAPI, React.js, PostgreSQL, MinIO, Scikit-learn

- Developed Full-stack platform featuring secure 2FA, personalized watchlists, and a hybrid recommendation engine with per-user caching, deployed on a scalable Kubernetes cluster with NGINX and MinIO.

Real-Time File Sharing & Chat Application | React.js, Node.js, Express.js, Socket.IO, MongoDB, GridFS

- Engineered a real-time chat application with JWT authentication, persistent group messaging, and scalable file sharing via GridFS, ensuring sub-500ms delivery with Socket.IO.

TECHNICAL SKILLS

Programming Languages: Python, C++, JavaScript, SQL, MATLAB, HTML/CSS

Web Development & APIs: FastAPI, React, Node.js, Express.js, WebSockets

Cloud & DevOps: AWS (S3, EC2, Lambda), Azure ML, Docker, Kubernetes, Git, CI/CD, Linux/Ubuntu

Databases: Mongoose, PostgreSQL, Microsoft SQL Server, DynamoDB, Firebase, MinIO

Machine Learning & AI: Pytorch, Scikit-learn, ONNX, CNN/CRNN, Transformers, AI, Deep Learning, NLP

Data Science Tools: Pandas, NumPy, SciPy, Matplotlib, Jupyter, OpenCV, NLTK